



Chapter 6

Decision Tree Learning

"Prediction is very difficult, especially if it's about the future."

— Niels Bohr

Decision Tree Learning is a widely used predictive model for supervised learning that spans over a number of practical applications in various areas. It is used for both classification and regression tasks. The decision tree model basically represents logical rules that predict the value of a target variable by inferring from data features. This chapter provides a keen insight into how to construct a decision tree and infer knowledge from the tree.

Learning Objectives

- Understand the structure of the decision tree
- Know about the fundamentals of Entropy
- Learn and understand popular univariate Decision Tree Induction algorithms such as ID3, C4.5, and multivariate decision tree algorithm such as CART
- Deal with continuous attributes using improved C4.5 algorithm
- Construct Classification and Regression Tree (CART) for classifying both categorical and continuous-valued target variables
- Construct regression trees where the target feature is a continuous-valued variable
- Understand the basics of validating and pruning of decision trees

6.1 INTRODUCTION TO DECISION TREE LEARNING MODEL

Decision tree learning model, one of the most popular supervised predictive learning models, classifies data instances with high accuracy and consistency. The model performs an *inductive inference* that reaches a general conclusion from observed examples. This model is variably used for solving complex classification applications.

Decision tree is a concept tree which summarizes the information contained in the training dataset in the form of a tree structure. Once the concept model is built, test data can be easily classified.

This model can be used to classify both categorical target variables and continuous-valued target variables. Given a training dataset X , this model computes a hypothesis function $f(X)$ as decision tree.

Inputs to the model are data instances or objects with a set of features or attributes which can be discrete or continuous and the output of the model is a decision tree which predicts or classifies the target class for the test data object.

In statistical terms, attributes or features are called as independent variables. The target feature or target class is called as response variable which indicates the category we need to predict on a test object.

The decision tree learning model generates a complete hypothesis space in the form of a tree structure with the given training dataset and allows us to search through the possible set of hypotheses which in fact would be a smaller decision tree as we walk through the tree. This kind of search bias is called as preference bias.

6.1.1 Structure of a Decision Tree

A decision tree has a structure that consists of a root node, internal nodes/decision nodes, branches, and terminal nodes/leaf nodes. The topmost node in the tree is the root node. Internal nodes are the test nodes and are also called as decision nodes. These nodes represent a choice or test of an input attribute and the outcome or outputs of the test condition are the branches emanating from this decision node. The branches are labelled as per the outcomes or output values of the test condition. Each branch represents a sub-tree or subsection of the entire tree. Every decision node is part of a path to a leaf node. The leaf nodes represent the labels or the outcome of a decision path. The labels of the leaf nodes are the different target classes a data instance can belong to.

Every path from root to a leaf node represents a logical rule that corresponds to a conjunction of test attributes and the whole tree represents a disjunction of these conjunctions. The decision tree model, in general, represents a collection of logical rules of classification in the form of a tree structure.

Decision networks, otherwise called as influence diagrams, have a directed graph structure with nodes and links. It is an extension of Bayesian belief networks that represents information about each node's current state, its possible actions, the possible outcome of those actions, and their utility. The concept of Bayesian Belief Network (BBN) is discussed in Chapter 9.

Figure 6.1 shows symbols that are used in this book to represent different nodes in the construction of a decision tree. A circle is used to represent a root node, a diamond symbol is used to represent a decision node or the internal nodes, and all leaf nodes are represented with a rectangle.

	Root node
	Decision node
	Leaf node

Figure 6.1: Nodes in a Decision Tree

A decision tree consists of two major procedures discussed below.

Building the Tree

Goal Construct a decision tree with the given training dataset. The tree is constructed in a top-down fashion. It starts from the root node. At every level of tree construction, we need to find the best split attribute or best decision node among all attributes. This process is recursive and continued until we end up in the last level of the tree or finding a leaf node which cannot be split further. The tree construction is complete when all the test conditions lead to a leaf node. The leaf node contains the target class or output of classification.

Output Decision tree representing the complete hypothesis space.

Knowledge Inference or Classification

Goal Given a test instance, infer to the target class it belongs to.

Classification Inferring the target class for the test instance or object is based on inductive inference on the constructed decision tree. In order to classify an object, we need to start traversing the tree from the root. We traverse as we evaluate the test condition on every decision node with the test object attribute value and walk to the branch corresponding to the test's outcome. This process is repeated until we end up in a leaf node which contains the target class of the test object.

Output Target label of the test instance.

Advantages of Decision Trees

1. Easy to model and interpret
2. Simple to understand
3. The input and output attributes can be discrete or continuous predictor variables.
4. Can model a high degree of nonlinearity in the relationship between the target variables and the predictor variables
5. Quick to train

Disadvantages of Decision Trees

Some of the issues that generally arise with a decision tree learning are that:

1. It is difficult to determine how deeply a decision tree can be grown or when to stop growing it.
2. If training data has errors or missing attribute values, then the decision tree constructed may become unstable or biased.
3. If the training data has continuous valued attributes, handling it is computationally complex and has to be discretized.
4. A complex decision tree may also be over-fitting with the training data.
5. Decision tree learning is not well suited for classifying multiple output classes.
6. Learning an optimal decision tree is also known to be NP-complete.

Example 6.1: How to draw a decision tree to predict a student's academic performance based on the given information such as class attendance, class assignments, home-work assignments, tests, participation in competitions or other events, group activities such as projects and presentations, etc.

Solution: The target feature is the student performance in the final examination whether he will pass or fail in the examination. The decision nodes are test nodes which check for conditions like 'What's the student's class attendance?', 'How did he perform in his class assignments?', 'Did he do his home assignments properly?', 'What about his assessment results?', 'Did he participate in competitions or other events?', 'What is the performance rating in group activities such as projects and presentations?'. Table 6.1 shows the attributes and set of values for each attribute.

Table 6.1: Attributes and Associated Values

Attributes	Values
Class attendance	Good, Average, Poor
Class assignments	Good, Moderate, Poor
Home-work assignments	Yes, No
Assessment	Good, Moderate, Poor
Participation in competitions or other events	Yes, No
Group activities such as projects and presentations	Yes, No
Exam Result	Pass, Fail

The leaf nodes represent the outcomes, that is, either 'pass', or 'fail'.

A decision tree would be constructed by following a set of if-else conditions which may or may not include all the attributes, and decision nodes outcomes are two or more than two. Hence, the tree is not a binary tree.

Note: A decision tree is not always a binary tree. It is a tree which can have more than two branches.

Example 6.2: Predict a student's academic performance of whether he will pass or fail based on the given information such as 'Assessment' and 'Assignment'. The following Table 6.2 shows the independent variables, Assessment and Assignment, and the target variable Exam Result with their values. Draw a binary decision tree.

Table 6.2: Attributes and Associated Values

Attributes	Values
Assessment	≥ 50 , < 50
Assignment	Yes, No
Exam Result	Pass, Fail

Solution: Consider the root node is 'Assessment'. If a student's marks are ≥ 50 , the root node is branched to leaf node 'Pass' and if the assessment marks are < 50 , it is branched to another decision node. If the decision node in next level of the tree is 'Assignment' and if a student has submitted his assignment, the node branches to 'Pass' and if not submitted, the node branches to 'Fail'. Figure 6.2 depicts this rule.

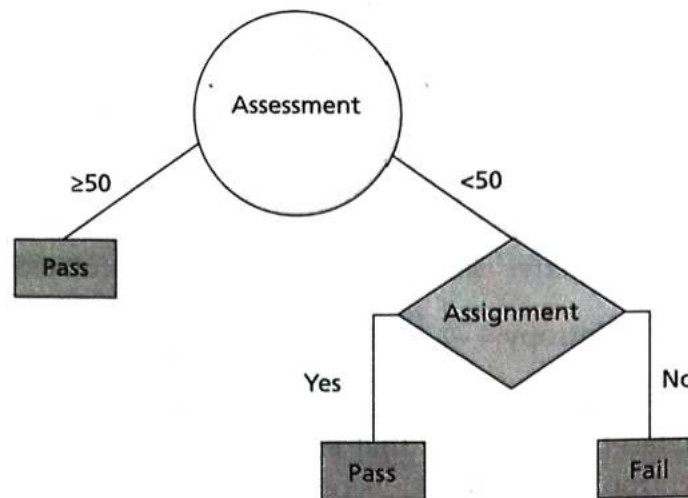


Figure 6.2: Illustration of a Decision Tree

This tree can be interpreted as a sequence of logical rules as follows:

if (Assessment ≥ 50) then 'Pass'

else if (Assessment < 50) then

 if (Assignment = Yes) then 'Pass'

 else if (Assignment = No) then 'Fail'

Now, if a test instance is given, such as a student has scored 42 marks in his assessment and has not submitted his assignment, then it is predicted with the decision tree that his exam result is 'Fail'.

Many algorithms exist which will be studied for constructing decision trees in the sections below.

6.1.2 Fundamentals of Entropy

Given the training dataset with a set of attributes or features, the decision tree is constructed by finding the attribute or feature that best describes the target class for the given test instances. The best split feature is the one which contains more information about how to split the dataset among all features so that the target class is accurately identified for the test instances. In other words, the best split attribute is more informative to split the dataset into sub datasets and this process is continued until the stopping criterion is reached. This splitting should be pure at every stage of selecting the best feature.

The best feature is selected based on the amount of information among the features which are basically calculated on probabilities. Quantifying information is closely related to information theory. In the field of information theory, the features are quantified by a measure called Shannon Entropy which is calculated based on the probability distribution of the events.

Entropy is the amount of uncertainty or randomness in the outcome of a random variable or an event. Moreover, entropy describes about the homogeneity of the data instances. The best feature is selected based on the entropy value. For example, when a coin is flipped, head or tail are the two outcomes, hence its entropy is lower when compared to rolling a dice which has got six outcomes. Hence, the interpretation is,

Example 6.1: How to draw a decision tree to predict a student's academic performance based on the given information such as class attendance, class assignments, home-work assignments, tests, participation in competitions or other events, group activities such as projects and presentations, etc.

Solution: The target feature is the student performance in the final examination whether he will pass or fail in the examination. The decision nodes are test nodes which check for conditions like 'What's the student's class attendance?', 'How did he perform in his class assignments?', 'Did he do his home assignments properly?', 'What about his assessment results?', 'Did he participate in competitions or other events?', 'What is the performance rating in group activities such as projects and presentations?'. Table 6.1 shows the attributes and set of values for each attribute.

Table 6.1: Attributes and Associated Values

Attributes	Values
Class attendance	Good, Average, Poor
Class assignments	Good, Moderate, Poor
Home-work assignments	Yes, No
Assessment	Good, Moderate, Poor
Participation in competitions or other events	Yes, No
Group activities such as projects and presentations	Yes, No
Exam Result	Pass, Fail

The leaf nodes represent the outcomes, that is, either 'pass', or 'fail'.

A decision tree would be constructed by following a set of if-else conditions which may or may not include all the attributes, and decision nodes outcomes are two or more than two. Hence, the tree is not a binary tree.

Note: A decision tree is not always a binary tree. It is a tree which can have more than two branches.

Example 6.2: Predict a student's academic performance of whether he will pass or fail based on the given information such as 'Assessment' and 'Assignment'. The following Table 6.2 shows the independent variables, Assessment and Assignment, and the target variable Exam Result with their values. Draw a binary decision tree.

Table 6.2: Attributes and Associated Values

Attributes	Values
Assessment	≥ 50 , < 50
Assignment	Yes, No
Exam Result	Pass, Fail

Solution: Consider the root node is 'Assessment'. If a student's marks are ≥ 50 , the root node is branched to leaf node 'Pass' and if the assessment marks are < 50 , it is branched to another decision node. If the decision node in next level of the tree is 'Assignment' and if a student has submitted his assignment, the node branches to 'Pass' and if not submitted, the node branches to 'Fail'. Figure 6.2 depicts this rule.

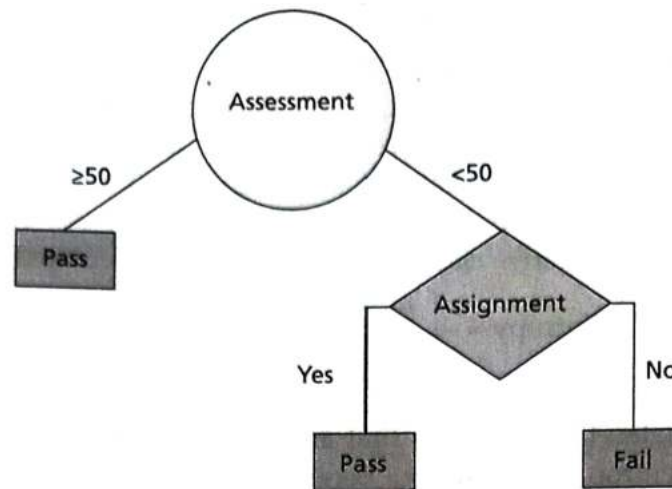


Figure 6.2: Illustration of a Decision Tree

This tree can be interpreted as a sequence of logical rules as follows:

if (Assessment ≥ 50) then 'Pass'

else if (Assessment < 50) then

 if (Assignment = Yes) then 'Pass'

 else if (Assignment = No) then 'Fail'

Now, if a test instance is given, such as a student has scored 42 marks in his assessment and has not submitted his assignment, then it is predicted with the decision tree that his exam result is 'Fail'.

Many algorithms exist which will be studied for constructing decision trees in the sections below.

6.1.2 Fundamentals of Entropy

Given the training dataset with a set of attributes or features, the decision tree is constructed by finding the attribute or feature that best describes the target class for the given test instances. The best split feature is the one which contains more information about how to split the dataset among all features so that the target class is accurately identified for the test instances. In other words, the best split attribute is more informative to split the dataset into sub datasets and this process is continued until the stopping criterion is reached. This splitting should be pure at every stage of selecting the best feature.

The best feature is selected based on the amount of information among the features which are basically calculated on probabilities. Quantifying information is closely related to information theory. In the field of information theory, the features are quantified by a measure called Shannon Entropy which is calculated based on the probability distribution of the events.

Entropy is the amount of uncertainty or randomness in the outcome of a random variable or an event. Moreover, entropy describes about the homogeneity of the data instances. The best feature is selected based on the entropy value. For example, when a coin is flipped, head or tail are the two outcomes, hence its entropy is lower when compared to rolling a dice which has got six outcomes. Hence, the interpretation is,

Higher the entropy → Higher the uncertainty

Lower the entropy → Lower the uncertainty

Similarly, if all instances are homogenous, say (1, 0), which means all instances belong to the same class (here it is positive) or (0, 1) where all instances are negative, then the entropy is 0. On the other hand, if the instances are equally distributed, say (0.5, 0.5), which means 50% positive and 50% negative, then the entropy is 1. If there are 10 data instances, out of which 6 belong to positive class and 4 belong to negative class, then the entropy is calculated as shown in Eq. (6.1),

$$\text{Entropy} = - \left[\frac{6}{10} \log_2 \frac{6}{10} + \frac{4}{10} \log_2 \frac{4}{10} \right] \quad (6.1)$$

It is concluded that if the dataset has instances that are completely homogeneous, then the entropy is 0 and if the dataset has samples that are equally divided (i.e., 50% – 50%), it has an entropy of 1. Thus, the entropy value ranges between 0 and 1 based on the randomness of the samples in the dataset. If the entropy is 0, then the split is pure which means that all samples in the set will partition into one class or category. But if the entropy is 1, the split is impure and the distribution of the samples is more random. The stopping criterion is based on the entropy value.

Let P be the probability distribution of data instances from 1 to n as shown in Eq. (6.2).

$$\text{So, } P = P_1, \dots, P_n \quad (6.2)$$

Entropy of P is the information measure of this probability distribution given in Eq. (6.3),

$$\begin{aligned} \text{Entropy_Info}(P) &= \text{Entropy_Info}(P_1, \dots, P_n) \\ &= -(P_1 \log_2(P_1) + P_2 \log_2(P_2) + \dots + P_n \log_2(P_n)) \end{aligned} \quad (6.3)$$

where, P_1 is the probability of data instances classified as class 1 and P_2 is the probability of data instances classified as class 2 and so on.

$P_1 = \text{No of data instances belonging to class 1} / \text{Total no of data instances in the training dataset}$

Entropy_Info(P) can be computed as shown in Eq. (6.4).

Thus,

$$\text{Entropy_Info}(6, 4) \text{ is calculated as } - \left[\frac{6}{10} \log_2 \frac{6}{10} + \frac{4}{10} \log_2 \frac{4}{10} \right] \quad (6.4)$$

Mathematically, entropy is defined in Eq. (6.5) as:

$$\text{Entropy_Info}(X) = \sum_{x \in \text{values}(X)} Pr[X = x] \cdot \log_2 \frac{1}{Pr[X = x]} \quad (6.5)$$

$Pr[X = x]$ is the probability of a random variable X with a possible outcome x .

Note: $\log_2 \frac{1}{Pr[X = x]} = -\log_2(Pr[X = x])$

Algorithm 6.1: General Algorithm for Decision Trees

1. Find the best attribute from the training dataset using an attribute selection measure and place it at the root of the tree.

(Continued)

2. Split the training dataset into subsets based on the outcomes of the test attribute and each subset in a branch contains the data instances or tuples with the same value for the selected test attribute.
3. Repeat step 1 and step 2 on each subset until we end up in leaf nodes in all the branches of the tree.
4. This splitting process is recursive until the stopping criterion is reached.

Stopping Criteria

The following are some of the common stopping conditions:

1. The data instances are homogenous which means all belong to the same class C_i and hence its entropy is 0.
2. A node with some defined minimum number of data instances becomes a leaf (Number of data instances in a node is between 0.25 and 1.00% of the full training dataset).
3. The maximum tree depth is reached, so further splitting is not done and the node becomes a leaf node.

6.2 DECISION TREE INDUCTION ALGORITHMS

There are many decision tree algorithms, such as ID3, C4.5, CART, CHAID, QUEST, GUIDE, CRUISE, and CTREE, that are used for classification in real-time environment. The most commonly used decision tree algorithms are ID3 (Iterative Dichotomizer 3), developed by J.R Quinlan in 1986, and C4.5 is an advancement of ID3 presented by the same author in 1993. CART, that stands for Classification and Regression Trees, is another algorithm which was developed by Breiman et al. in 1984.

The accuracy of the tree constructed depends upon the selection of the best split attribute. Different algorithms are used for building decision trees which use different measures to decide on the splitting criterion. Algorithms such as ID3, C4.5 and CART are popular algorithms used in the construction of decision trees. The algorithm ID3 uses 'Information Gain' as the splitting criterion whereas the algorithm C4.5 uses 'Gain Ratio' as the splitting criterion. The CART algorithm is popularly used for classifying both categorical and continuous-valued target variables. CART uses GINI Index to construct a decision tree.

Decision trees constructed using ID3 and C4.5 are also called as *univariate decision trees* which consider only one feature/attribute to split at each decision node whereas decision trees constructed using CART algorithm are *multivariate decision trees* which consider a conjunction of univariate splits. The details about univariate and multivariate data has been discussed in Chapter 2.

6.2.1 ID3 Tree Construction

ID3 is a supervised learning algorithm which uses a training dataset with labels and constructs a decision tree. ID3 is an example of univariate decision trees as it considers only one feature at each decision node. This leads to axis-aligned splits. The tree is then used to classify the future test instances. It constructs the tree using a greedy approach in a top-down fashion by identifying the best attribute at each level of the tree.

ID3 works well if the attributes or features are considered as discrete/categorical values. If some attributes are continuous, then we have to partition attributes or features to be discretized or nominal attributes or features.

The algorithm builds the tree using a purity measure called 'Information Gain' with the given training data instances and then uses the constructed tree to classify the test data. It is applied for training set with only nominal attributes or categorical attributes and with no missing values for classification. ID3 works well for a large dataset. If the dataset is small, overfitting may occur. Moreover, it is not accurate if the dataset has missing attribute values.

No pruning is done during or after construction of the tree and it is prone to outliers. C4.5 and CART can handle both categorical attributes and continuous attributes. Both C4.5 and CART can also handle missing values, but C4.5 is prone to outliers whereas CART can handle outliers as well.

Algorithm 6.2: Procedure to Construct a Decision Tree using ID3

1. Compute Entropy_Info Eq. (6.8) for the whole training dataset based on the target attribute.
2. Compute Entropy_Info Eq. (6.9) and Information_Gain Eq. (6.10) for each of the attribute in the training dataset.
3. Choose the attribute for which entropy is minimum and therefore the gain is maximum as the best split attribute.
4. The best split attribute is placed as the root node.
5. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into subsets.
6. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

Note: We stop branching a node if entropy is 0.

The best split attribute at every iteration is the attribute with the highest information gain.

Definitions

Let T be the training dataset.

Let A be the set of attributes $A = \{A_1, A_2, A_3, \dots, A_n\}$.

Let m be the number of classes in the training dataset.

Let P_i be the probability that a data instance or a tuple ' d ' belongs to class C_i .

It is calculated as,

$P_i = \frac{\text{Total no of data instances that belongs to class } C_i \text{ in } T}{\text{Total no of tuples in the training set } T}$ (6.6)

Mathematically, it is represented as shown in Eq. (6.7).

$$P_i = \frac{|d_{C_i}|}{|T|} \quad (6.7)$$

Expected information or Entropy needed to classify a data instance d' in T is denoted as $\text{Entropy_Info}(T)$ given in Eq. (6.8).

$$\text{Entropy_Info}(T) = - \sum_{i=1}^m P_i \log_2 P_i \quad (6.8)$$

Entropy of every attribute denoted as $\text{Entropy_Info}(T, A)$ is shown in Eq. (6.9) as:

$$\text{Entropy_Info}(T, A) = \sum_{i=1}^v \frac{|A_i|}{|T|} \times \text{Entropy_Info}(A_i) \quad (6.9)$$

where, the attribute A has got ' v ' distinct values $\{a_1, a_2, \dots, a_v\}$, $|A_i|$ is the number of instances for distinct value ' i ' in attribute A , and $\text{Entropy_Info}(A_i)$ is the entropy for that set of instances.

Information_Gain is a metric that measures how much information is gained by branching on an attribute A . In other words, it measures the reduction in impurity in an arbitrary subset of data.

It is calculated as given in Eq. (6.10):

$$\text{Information_Gain}(A) = \text{Entropy_Info}(T) - \text{Entropy_Info}(T, A) \quad (6.10)$$

It can be noted that as entropy increases, information gain decreases. They are inversely proportional to each other.

Scan for 'Additional Examples'



Example 6.3: Assess a student's performance during his course of study and predict whether a student will get a job offer or not in his final year of the course. The training dataset T consists of 10 data instances with attributes such as 'CGPA', 'Interactiveness', 'Practical Knowledge' and 'Communication Skills' as shown in Table 6.3. The target class attribute is the 'Job Offer'.

Table 6.3: Training Dataset T

S.No.	CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
1.	≥ 9	Yes	Very good	Good	Yes
2.	≥ 8	No	Good	Moderate	Yes
3.	≥ 9	No	Average	Poor	No
4.	< 8	No	Average	Good	No
5.	≥ 8	Yes	Good	Moderate	Yes
6.	≥ 9	Yes	Good	Moderate	Yes
7.	< 8	Yes	Good	Poor	No
8.	≥ 9	No	Very good	Good	Yes
9.	≥ 8	Yes	Good	Good	Yes
10.	≥ 8	Yes	Average	Good	Yes

Solution:**Step 1:**

Calculate the Entropy for the target class 'Job Offer'.

$$\begin{aligned} \text{Entropy_Info}(\text{Target Attribute} = \text{Job Offer}) &= \text{Entropy_Info}(7, 3) = \\ &= -\left[\frac{7}{10} \log_2 \frac{7}{10} + \frac{3}{10} \log_2 \frac{3}{10} \right] = -(-0.3599 + -0.5208) = 0.8807 \end{aligned}$$

Iteration 1:**Step 2:**

Calculate the Entropy_Info and Gain(Information_Gain) for each of the attribute in the training dataset.

Table 6.4 shows the number of data instances classified with Job Offer as Yes or No for the attribute CGPA.

Table 6.4: Entropy Information for CGPA

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
≥9	3	1	4	
≥8	4	0	4	0
<8	0	2	2	0

Entropy_Info(T, CGPA)

$$\begin{aligned} &= \frac{4}{10} \left[-\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right] + \frac{4}{10} \left[-\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} \right] + \frac{2}{10} \left[-\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} \right] \\ &= \frac{4}{10} (0.3111 + 0.4997) + 0 + 0 \\ &= 0.3243 \end{aligned}$$

$$\begin{aligned} \text{Gain (CGPA)} &= 0.8807 - 0.3243 \\ &= 0.5564 \end{aligned}$$

Table 6.5 shows the number of data instances classified with Job Offer as Yes or No for the attribute Interactiveness.

Table 6.5: Entropy Information for Interactiveness

Interactiveness	Job Offer = Yes	Job Offer = No	Total	Entropy
YES	5	1	6	
NO	2	2	4	

$$\begin{aligned} \text{Entropy_Info}(T, \text{Interactiveness}) &= \frac{6}{10} \left[-\frac{5}{6} \log_2 \frac{5}{6} - \frac{1}{6} \log_2 \frac{1}{6} \right] + \frac{4}{10} \left[-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \right] \\ &= \frac{6}{10} (0.2191 + 0.4306) + \frac{4}{10} (0.4997 + 0.4997) \\ &= 0.3898 + 0.3998 = 0.7896 \end{aligned}$$

$$\begin{aligned} \text{Gain(Interactiveness)} &= 0.8807 - 0.7896 \\ &= 0.0911 \end{aligned}$$

Table 6.6 shows the number of data instances classified with Job Offer as Yes or No for the attribute Practical Knowledge.

Table 6.6: Entropy Information for Practical Knowledge

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Very Good	2	0	2	0
Average	1	2	3	
Good	4	1	5	

Entropy_Info(T , Practical Knowledge)

$$\begin{aligned}
 &= \frac{2}{10} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{3}{10} \left[-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \right] + \frac{5}{10} \left[-\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} \right] \\
 &= \frac{2}{10} (0) + \frac{3}{10} (0.5280 + 0.3897) + \frac{5}{10} (0.2574 + 0.4641) \\
 &= 0 + 0.2753 + 0.3608 \\
 &= 0.6361
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain(Practical Knowledge)} &= 0.8807 - 0.6361 \\
 &= 0.2446
 \end{aligned}$$

Table 6.7 shows the number of data instances classified with Job Offer as Yes or No for the attribute Communication Skills.

Table 6.7: Entropy Information for Communication Skills

Communication Skills	Job Offer = Yes	Job Offer = No	Total
Good	4	1	5
Moderate	3	0	3
Poor	0	2	2

Entropy_Info(T , Communication Skills)

$$\begin{aligned}
 &= \frac{5}{10} \left[-\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} \right] + \frac{3}{10} \left[-\frac{3}{3} \log_2 \frac{3}{3} - \frac{0}{3} \log_2 \frac{0}{3} \right] + \frac{2}{10} \left[-\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} \right] \\
 &= \frac{5}{10} (0.5280 + 0.3897) + \frac{3}{10} (0) + \frac{2}{10} (0) \\
 &= 0.3609
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain(Communication Skills)} &= 0.8813 - 0.36096 \\
 &= 0.5203
 \end{aligned}$$

The Gain calculated for all the attributes is shown in Table 6.8:

Table 6.8: Gain

Attributes	Gain
CGPA	0.5564
Interactiveness	0.0911
Practical Knowledge	0.2246
Communication Skills	0.5203

Step 3: From Table 6.8, choose the attribute for which entropy is minimum and therefore the gain is maximum as the best split attribute.

The best split attribute is CGPA since it has the maximum gain. So, we choose CGPA as the root node. There are three distinct values for CGPA with outcomes ≥ 9 , ≥ 8 and < 8 . The entropy value is 0 for ≥ 8 and < 8 with all instances classified as Job Offer = Yes for ≥ 8 and Job Offer = No for < 8 . Hence, both ≥ 8 and < 8 end up in a leaf node. The tree grows with the subset of instances with CGPA ≥ 9 as shown in Figure 6.3.

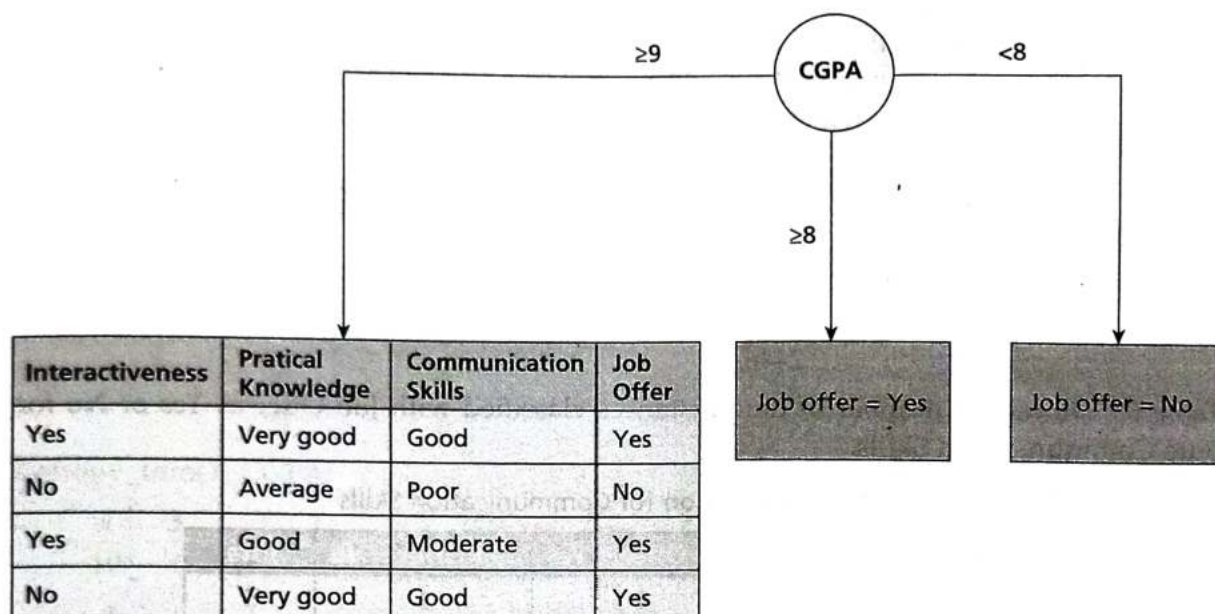


Figure 6.3: Decision Tree After Iteration 1

Now, continue the same process for the subset of data instances branched with CGPA ≥ 9 .

Iteration 2:

In this iteration, the same process of computing the Entropy_Info and Gain are repeated with the subset of training set. The subset consists of 4 data instances as shown in the above Figure 6.3.

$$\text{Entropy_Info}(T) = \text{Entropy_Info}(3, 1) =$$

$$= -\left[\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right]$$

$$= -(-0.3111 + -0.4997)$$

$$= 0.8108$$

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{2}{4} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{2}{4} \left[-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right]$$

$$= 0 + 0.4997$$

$$\text{Gain}(\text{Interactiveness}) = 0.8108 - 0.4997$$

$$= 0.3111$$

$$\text{Entropy_Info}(T, \text{Practical Knowledge})$$

$$= \frac{2}{4} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{1}{4} \left[-\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \right] + \frac{1}{4} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right]$$

$$= 0$$

$$\text{Gain}(\text{Practical Knowledge}) = 0.8108$$

$$\text{Entropy_Info}(T, \text{Communication Skills})$$

$$= \frac{2}{4} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{1}{4} \left[-\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \right] + \frac{1}{4} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right]$$

$$= 0$$

$$\text{Gain}(\text{Communication Skills}) = 0.8108$$

The gain calculated for all the attributes is shown in Table 6.9.

Table 6.9: Total Gain

Attributes	Gain
Interactiveness	0.3111
Practical Knowledge	0.8108
Communication Skills	0.8108

Here, both the attributes 'Practical Knowledge' and 'Communication Skills' have the same Gain. So, we can either construct the decision tree using 'Practical Knowledge' or 'Communication Skills'. The final decision tree is shown in Figure 6.4.

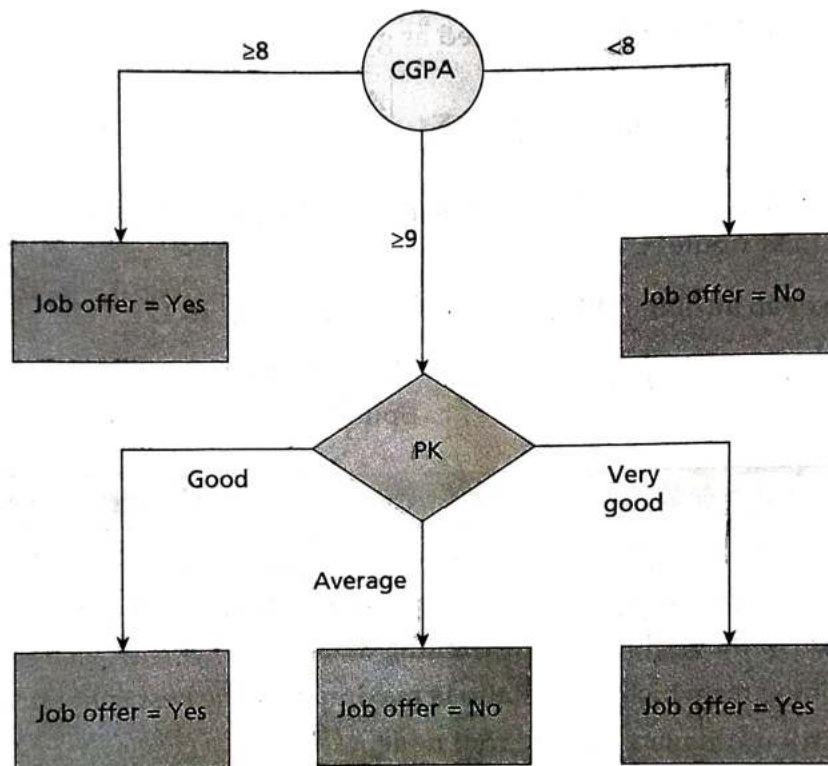


Figure 6.4: Final Decision Tree

6.2.2 C4.5 Construction

C4.5 is an improvement over ID3. C4.5 works with continuous and discrete attributes and missing values, and it also supports post-pruning. C5.0 is the successor of C4.5 and is more efficient and used for building smaller decision trees. C4.5 works with missing values by marking as '?', but these missing attribute values are not considered in the calculations.

The algorithm C4.5 is based on Occam's Razor which says that given two correct solutions, the simpler solution has to be chosen. Moreover, the algorithm requires a larger training set for better accuracy. It uses Gain Ratio as a measure during the construction of decision trees. ID3 is more biased towards attributes with larger values. For example, if there is an attribute called 'Register No' for students it would be unique for every student and will have distinct value for every data instance resulting in more values for the attribute. Hence, every instance belongs to a category and would have higher Information Gain than other attributes. To overcome this bias issue, C4.5 uses a purity measure Gain ratio to identify the best split attribute. In C4.5 algorithm, the Information Gain measure used in ID3 algorithm is normalized by computing another factor called Split_Info. This normalized information gain of an attribute called as Gain_Ratio is computed by the ratio of the calculated Split_Info and Information Gain of each attribute. Then, the attribute with the highest normalized information gain, that is, highest gain ratio is used as the splitting criteria.

As an example, we will choose the same training dataset shown in Table 6.3 to construct a decision tree using the C4.5 algorithm.

Given a Training dataset T ,

The Split_Info of an attribute A is computed as given in Eq. (6.11):

$$\text{Split_Info}(T, A) = -\sum_{i=1}^v \frac{|A_i|}{|T|} \times \log_2 \frac{|A_i|}{|T|} \quad (6.11)$$

where, the attribute A has got ' v ' distinct values $\{a_1, a_2, \dots, a_v\}$, and $|A_i|$ is the number of instance for distinct value ' i ' in attribute A .

The Gain_Ratio of an attribute A is computed as given in Eq. (6.12):

$$\text{Gain_Ratio}(A) = \frac{\text{Info_Gain}(A)}{\text{Split_Info}(T, A)}$$

Algorithm 6.3: Procedure to Construct a Decision Tree using C4.5

1. Compute Entropy_Info Eq. (6.8) for the whole training dataset based on the target attribute.
2. Compute Entropy_Info Eq. (6.9), Info_Gain Eq. (6.10), Split_Info Eq. (6.11) and Gain_Ratio Eq. (6.12) for each of the attribute in the training dataset.
3. Choose the attribute for which Gain_Ratio is maximum as the best split attribute.
4. The best split attribute is placed as the root node.
5. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into subsets.
6. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

Example 6.4: Make use of Information Gain of the attributes which are calculated in ID3 algorithm in Example 6.3 to construct a decision tree using C4.5.

Solution:

Iteration 1:

Step 1: Calculate the Class_Entropy for the target class 'Job Offer'.

$$\begin{aligned}\text{Entropy_Info}(\text{Target Attribute} = \text{Job Offer}) &= \text{Entropy_Info}(7, 3) = \\ &= -\left[\frac{7}{10} \log_2 \frac{7}{10} + \frac{3}{10} \log_2 \frac{3}{10}\right] \\ &= (-0.3599 + -0.5208) \\ &= 0.8807\end{aligned}$$

Step 2: Calculate the Entropy_Info, Gain(Info_Gain), Split_Info, Gain_Ratio for each of the attribute in the training dataset.

CGPA:

$$\begin{aligned}\text{Entropy_Info}(T, \text{CGPA}) &= \frac{4}{10} \left[-\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right] + \frac{4}{10} \left[-\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} \right] \\ &\quad + \frac{2}{10} \left[-\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} \right] \\ &= \frac{4}{10} (0.3111 + 0.4997) + 0 + 0 \\ &= 0.3243 \\ \text{Gain}(\text{CGPA}) &= 0.8807 - 0.3243 \\ &= 0.5564\end{aligned}$$

$$\begin{aligned}\text{Split_Info}(T, \text{CGPA}) &= -\frac{4}{10} \log_2 \frac{4}{10} - \frac{4}{10} \log_2 \frac{4}{10} - \frac{2}{10} \log_2 \frac{2}{10} \\ &= 0.5285 + 0.5285 + 0.4641 \\ &= 1.5211\end{aligned}$$

$$\begin{aligned}\text{Gain_Ratio}(\text{CGPA}) &= (\text{Gain}(\text{CGPA})) / (\text{Split_Info}(T, \text{CGPA})) \\ &= \frac{0.5564}{1.5211} = 0.3658\end{aligned}$$

Interactiveness:

$$\begin{aligned}\text{Entropy_Info}(T, \text{Interactiveness}) &= \frac{6}{10} \left[-\frac{5}{6} \log_2 \frac{5}{6} - \frac{1}{6} \log_2 \frac{1}{6} \right] + \frac{4}{10} \left[-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \right] \\ &= \frac{6}{10} (0.2191 + 0.4306) + \frac{4}{10} (0.4997 + 0.4997) \\ &= 0.3898 + 0.3998 = 0.7896\end{aligned}$$

$$\text{Gain}(\text{Interactiveness}) = 0.8807 - 0.7896 = 0.0911$$

$$\text{Gain}(\text{Interactiveness}) = -\frac{6}{10} \log_2 \frac{6}{10} - \frac{4}{10} \log_2 \frac{4}{10} = 0.9704$$

$$\begin{aligned}
 \text{Gain_Ratio(Interactiveness)} &= \frac{\text{Gain(Interactiveness)}}{\text{Split_Info}(T, \text{Interactiveness})} \\
 &= \frac{0.0911}{0.9704} \\
 &= 0.0939
 \end{aligned}$$

Practical Knowledge:

$$\begin{aligned}
 \text{Entropy_Info}(T, \text{Practical Knowledge}) &= \frac{2}{10} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{3}{10} \left[-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \right] \\
 &\quad + \frac{5}{10} \left[-\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} \right] \\
 &= \frac{2}{10}(0) + \frac{3}{10}(0.5280 + 0.3897) + \frac{5}{10}(0.2574 + 0.4641) \\
 &= 0 + 0.2753 + 0.3608 = 0.6361
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain(Practical Knowledge)} &= 0.8807 - 0.6361 \\
 &= 0.2448
 \end{aligned}$$

$$\begin{aligned}
 \text{Split_Info}(T, \text{Practical Knowledge}) &= -\frac{2}{10} \log_2 \frac{2}{10} - \frac{5}{10} \log_2 \frac{5}{10} - \frac{3}{10} \log_2 \frac{3}{10} \\
 &= 1.4853
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain_Ratio(Practical Knowledge)} &= \frac{\text{Gain(Practical Knowledge)}}{\text{Split_Info}(T, \text{Practical Knowledge})} \\
 &= \frac{0.2448}{1.4853} \\
 &= 0.1648
 \end{aligned}$$

Communication Skills:

$$\begin{aligned}
 \text{Entropy_Info}(T, \text{Communication Skills}) &= \frac{5}{10} \left[-\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} \right] + \frac{3}{10} \left[-\frac{3}{3} \log_2 \frac{3}{3} - \frac{0}{3} \log_2 \frac{0}{3} \right] \\
 &\quad + \frac{2}{10} \left[-\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} \right] \\
 &= \frac{5}{10}(0.5280 + 0.3897) + \frac{3}{10}(0) + \frac{2}{10}(0) \\
 &= 0.3609
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain(Communication Skills)} &= 0.8813 - 0.36096 \\
 &= 0.5202
 \end{aligned}$$

$$\begin{aligned}
 \text{Split_Info}(T, \text{Communication Skills}) &= -\frac{5}{10} \log_2 \frac{5}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{2}{10} \log_2 \frac{2}{10} \\
 &= 1.4853
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain_Ratio(Communication Skills)} &= \frac{\text{Gain(Communication Skills)}}{\text{Split_Info}(T, \text{Communication Skills})} \\
 &= \frac{0.5202}{1.4853} = 0.3502
 \end{aligned}$$

Table 6.10 shows the Gain_Ratio computed for all the attributes.

Table 6.10: Gain_Ratio

Attribute	Gain_Ratio
CGPA	0.3658
INTERACTIVENESS	0.0939
PRACTICAL KNOWLEDGE	0.1648
COMMUNICATION SKILLS	0.3502

Step 3: Choose the attribute for which Gain_Ratio is maximum as the best split attribute.

From Table 6.10, we can see that CGPA has highest gain ratio and it is selected as the best split attribute. We can construct the decision tree placing CGPA as the root node shown in Figure 6.5. The training dataset is split into subsets with 4 data instances.

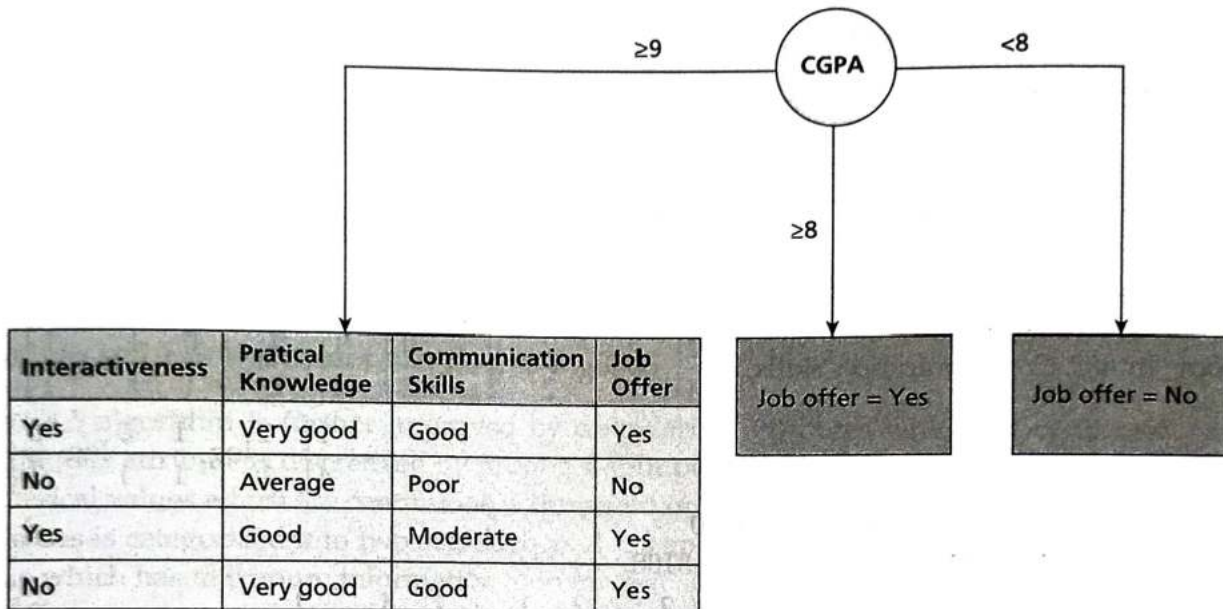


Figure 6.5: Decision Tree after Iteration 1

Iteration 2:

Total Samples: 4

Repeat the same process for this resultant dataset with 4 data instances.

Job Offer has 3 instances as Yes and 1 instance as No.

$$\begin{aligned}
 \text{Entropy_Info}(\text{Target Class} = \text{Job Offer}) &= -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \\
 &= 0.3112 + 0.5 \\
 &= 0.8112
 \end{aligned}$$

Interactiveness:

$$\begin{aligned}
 \text{Entropy_Info}(T, \text{Interactiveness}) &= \frac{2}{4} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{2}{4} \left[-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right] \\
 &= 0 + 0.4997
 \end{aligned}$$

$$\text{Gain}(\text{Interactiveness}) = 0.8108 - 0.4997 = 0.3111$$

$$\text{Split_Info}(T, \text{Interactiveness}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 0.5 + 0.5 = 1$$

$$\begin{aligned} \text{Gain_Ratio}(\text{Interactiveness}) &= \frac{\text{Gain}(\text{Interactiveness})}{\text{Split_Info}(T, \text{Interactiveness})} \\ &= \frac{0.3112}{1} = 0.3112 \end{aligned}$$

Practical Knowledge:

$$\begin{aligned} \text{Entropy_Info}(T, \text{Practical Knowledge}) &= \frac{2}{4} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{1}{4} \left[-\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \right] \\ &\quad + \frac{1}{4} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right] \\ &= 0 \end{aligned}$$

$$\text{Gain}(\text{Practical Knowledge}) = 0.8108$$

$$\text{Split_Info}(T, \text{Practical Knowledge}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 1.5$$

$$\text{Gain_Ratio}(\text{Practical Knowledge}) = \frac{\text{Gain}(\text{Practical Knowledge})}{\text{Split_Info}(T, \text{Practical Knowledge})} = \frac{0.8108}{1.5} = 0.5408$$

Communication Skills:

$$\begin{aligned} \text{Entropy_Info}(T, \text{Communication Skills}) &= \frac{2}{4} \left[-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right] + \frac{1}{4} \left[-\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \right] \\ &\quad + \frac{1}{4} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right] \\ &= 0 \end{aligned}$$

$$\text{Gain}(\text{Communication Skills}) = 0.8108$$

$$\text{Split_Info}(T, \text{Communication Skills}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 1.5$$

$$\text{Gain_Ratio}(\text{Communication Skills}) = \frac{\text{Gain}(\text{Practical Knowledge})}{\text{Split_Info}(T, \text{Practical Knowledge})} = \frac{0.8108}{1.5} = 0.5408$$

Table 6.11 shows the Gain_Ratio computed for all the attributes.

Table 6.11: Gain-Ratio

Attributes	Gain_Ratio
Interactiveness	0.3112
Practical Knowledge	0.5408
Communication Skills	0.5408

Both 'Practical Knowledge' and 'Communication Skills' have the highest gain ratio. So, the best splitting attribute can either be 'Practical Knowledge' or 'Communication Skills', and therefore, the split can be based on any one of these.

Here, we split based on 'Practical Knowledge'. The final decision tree is shown in Figure 6.6.

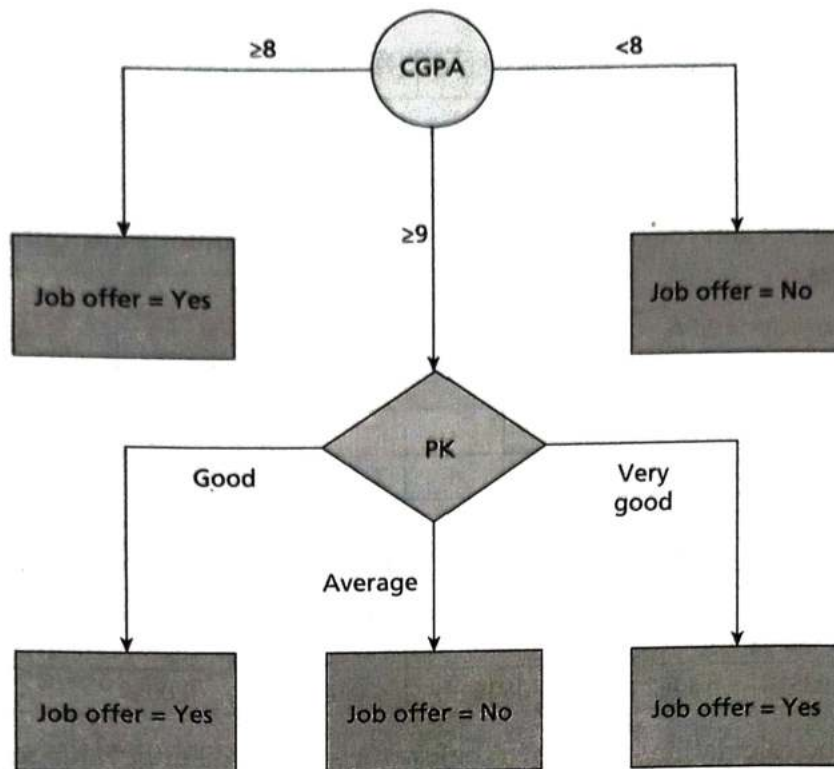


Figure 6.6: Final Decision Tree

Dealing with Continuous Attributes in C4.5

The C4.5 algorithm is further improved by considering attributes which are continuous, and a continuous attribute is discretized by finding a split point or threshold. When an attribute 'A' has numerical values which are continuous, a threshold or best split point 's' is found such that the set of values is categorized into two sets such as $A < s$ and $A \geq s$. The best split point is the attribute value which has maximum information gain for that attribute.

Now, let us consider the set of continuous values for the attribute CGPA in the sample dataset as shown in Table 6.12.

Table 6.12: Sample Dataset

S.No.	CGPA	Job Offer
1.	9.5	Yes
2.	8.2	Yes
3.	9.1	No
4.	6.8	No
5.	8.5	Yes
6.	9.5	Yes
7.	7.9	No
8.	9.1	Yes
9.	8.8	Yes
10.	8.8	Yes